National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	**Digital Signal Processing Lab**	**Course Code:**	**EL3031**
	Program:	**BS Electrical Engineering**	**Semester:**	**Spring 2025**
	Submission Date:	**9th-May 2025**	**Total Marks:**	**60 [Demonstration 40 + report and quiz (20)]**
	Section:	**BEE-6A**	**Page(s):**	**2**
	Exam Type:	**Design Project**	**LLO #**	**3, 4**

Problem statement:

As an Engineer, you are assigned the responsibility to work on a device, which effectively eliminates the high-frequency noise from real-life measurement signals obtained from a Gyroscope sensor of your mobile phone. The system must apply each of the following signal processing techniques explicitly in MATLAB for noise suppression:

i) Discrete Fourier Transform
ii) Butterworth Low Pass Filter
iii) Chebyshev Type I Low Pass Filter
iv) Chebyshev Type II Low Pass Filter
v) Elliptic Low Pass Filter

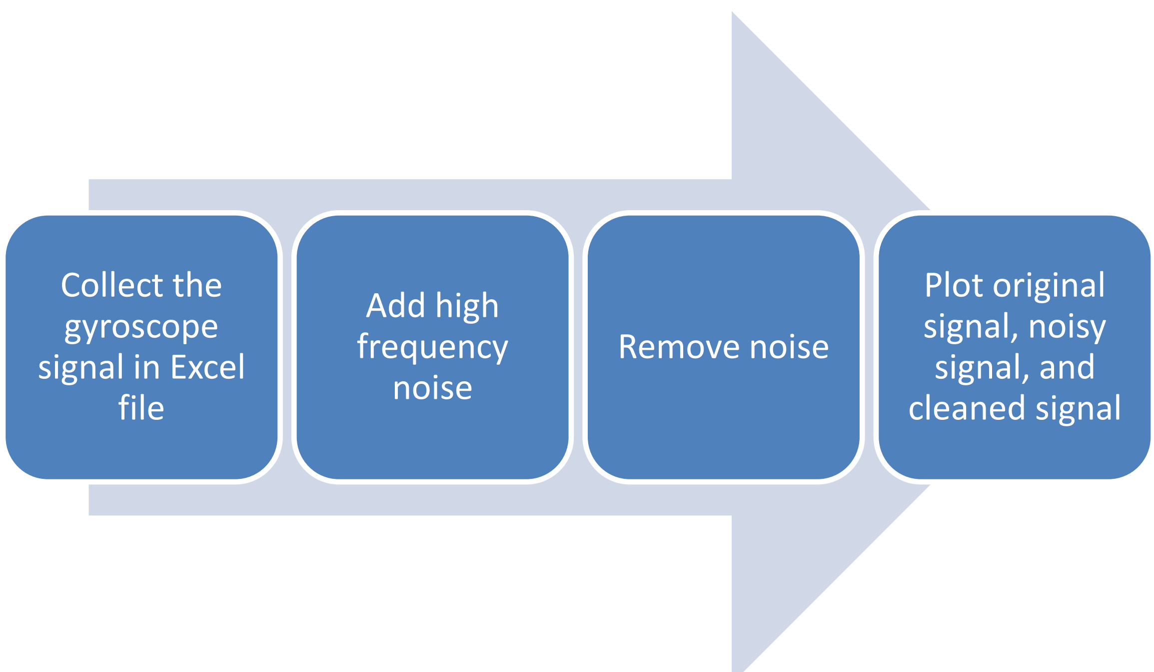

Figure 1: Flow diagram of the project

Guidelines for Collecting Gyroscope Data:

1. Download and install the "Physics Toolbox Sensor Suite" app on your mobile device. (Refer to the logo given in Fig. 2 for correct identification.)
2. After installation, open the app and select Gyroscope from the menu located at the upper left corner of the screen.
3. Press the Record button to start data collection.
4. **Recording Protocol:**
   Repeat the following sequence 10 times:

Move the mobile phone once, then remain still for 2 seconds. By repeating this pattern ten times, your recorded signal should resemble the signal shown in Fig. 3.

5. After completing the recording, stop the recording session and save the file in .CSV format.
6. Import the saved .CSV file into MATLAB, and proceed with the next steps according to the provided flow diagram (as shown in Fig. 1) to continue the project.

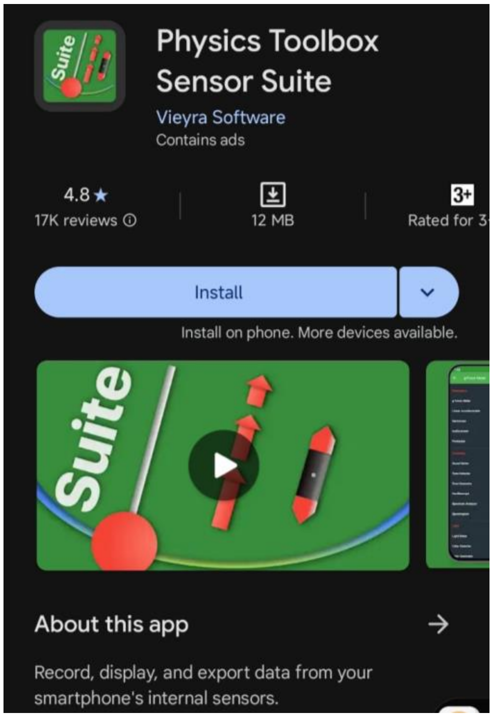

Figure 2: Logo of application

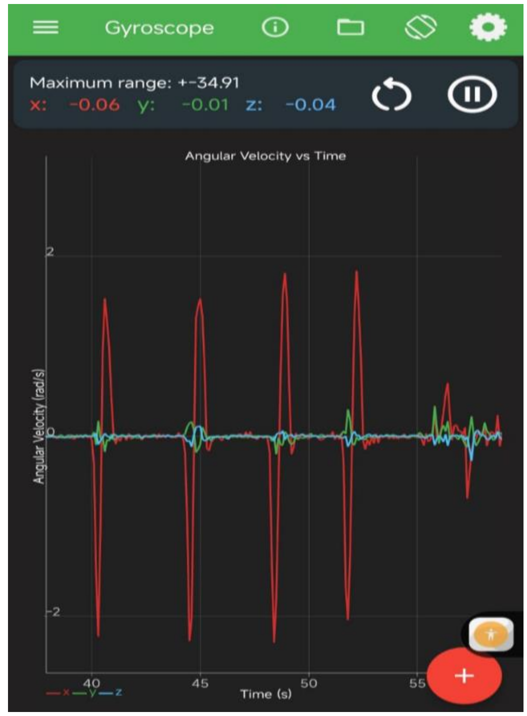

Figure 3: Raw signal of Gyroscope

Project Deliverables:

1. Report (in .pdf format and hard copy)
2. Contents of report should include: Introduction, Problem Analysis, Design requirements, possible solutions, Design description, Software simulations, Experimental results, Performance analysis, Future scope, and conclusion.
3. Code should be submitted separately.

Submission Details:

- You must do this assignment in a group of 2 students (maximum).
- Project Quiz to be conducted in the lab session.

Note:

- Make necessary assumptions where required.
- Submit efficient and properly commented code.
- Academic integrity is expected of all the students. Plagiarism or cheating in any assessment will result in negative marking or an **F** grade in the course.